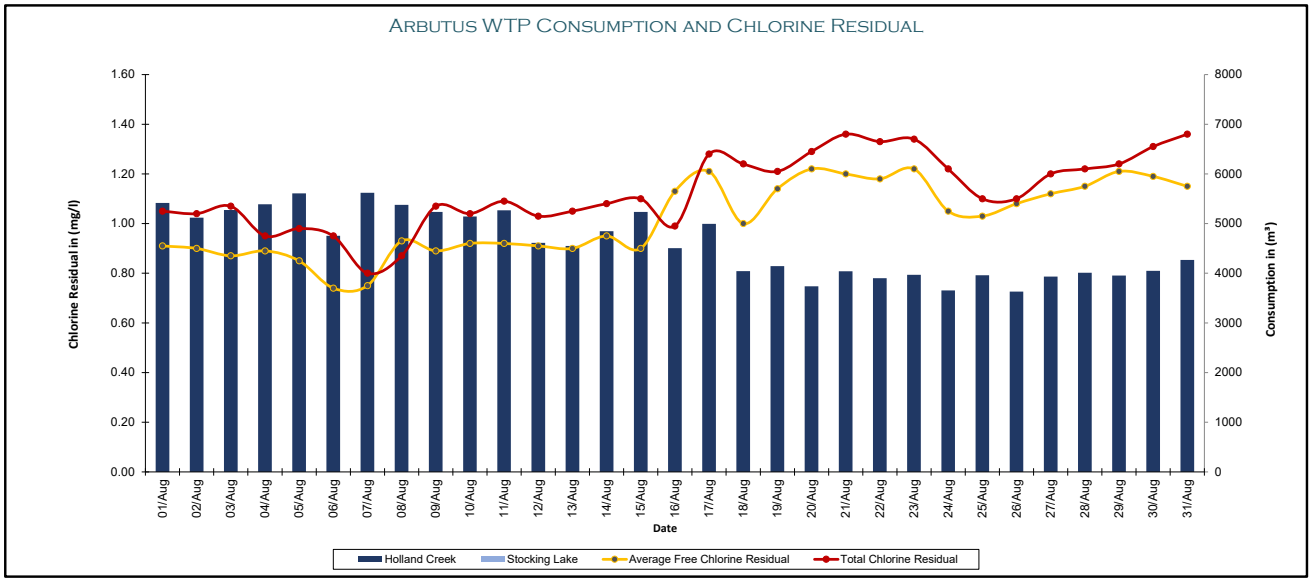
Town of Ladysmith - Arbutus Water treatment Plant

August 2024 - Monthly Report

Date	Daily Flow			Chlorine Residual				CT*	External Lab Testing					
	Stocking Lake	Holland Creek	Combined Flow	Free Min	Free Max	Free Avg	Total		HPC	E.coli	Total Coliforms	Aluminum	THM	HAA
	m³	m³	m³	mg/l	mg/l	mg/l	mg/l	Minutes·mg/l	CFU	MPN	MPN	mg/l	mg/l	mg/l
01-Aug	0	5417	5417	0.88	0.92	0.91	1.05	107						
02-Aug	0	5120	5120	0.88	0.92	0.90	1.04	141						
03-Aug	0	5273	5273	0.85	0.91	0.87	1.07	158						
04-Aug	0	5389	5389	0.79	0.90	0.89	0.95	203						
05-Aug	0	5607	5607	0.83	0.90	0.85	0.98	143	< 1	< 1	< 1			
06-Aug	0	4754	4754	0.71	0.87	0.74	0.95	152						
07-Aug	0	5619	5619	0.70	0.76	0.75	0.80	166						
08-Aug	0	5376	5376	0.72	0.93	0.93	0.87	326						
09-Aug	0	5234	5234	0.86	1.00	0.89	1.07	182						
10-Aug	0	5143	5143	0.87	0.93	0.92	1.04	204						
11-Aug	0	5266	5266	0.90	0.95	0.92	1.09	217						
12-Aug	0	4613	4613	0.87	0.93	0.91	1.03	215						
13-Aug	0	4550	4550	0.86	0.92	0.90	1.05	202	< 1	< 1	< 1		0.0733	0.0665
14-Aug	0	4847	4847	0.88	0.96	0.95	1.08	189						
15-Aug	0	5235	5235	0.88	0.98	0.90	1.10	161						
16-Aug	0	4504	4504	0.85	1.13	1.13	0.99	220						
17-Aug	0	4994	4994	1.12	1.24	1.21	1.28	192						
18-Aug	0	4042	4042	1.00	1.23	1.00	1.24	255						
19-Aug	0	4144	4144	0.96	1.15	1.14	1.21	288						
20-Aug	0	3738	3738	1.11	1.26	1.22	1.29	290	< 1	< 1	< 1		0.0876	0.0724
21-Aug	0	4040	4040	1.12	1.22	1.20	1.36	303						
22-Aug	0	3900	3900	1.15	1.20	1.18	1.33	255						
23-Aug	0	3969	3969	1.16	1.23	1.22	1.34	258						
24-Aug	0	3654	3654	1.05	1.22	1.05	1.22	238						
25-Aug	0	3961	3961	0.98	1.04	1.03	1.10	257						
26-Aug	0	3632	3632	1.00	1.09	1.08	1.10	257						
27-Aug	0	3933	3933	1.06	1.13	1.12	1.20	250	< 1	< 1	< 1		0.0819	0.0585
28-Aug	0	4011	4011	1.07	1.15	1.15	1.22	263						
29-Aug	0	3955	3955	1.14	1.22	1.21	1.24	253						
30-Aug	0	4049	4049	1.18	1.23	1.19	1.31	254						
31-Aug	0	4269	4269	1.15	1.20	1.15	1.36	241						
Total	0	142238	142238											
Average	0	4588	4588	0.95	1.06	1.02	1.13	221	< 1	< 1	< 1	#DIV/0!	0.0809	0.06580

*CT - Recorded as the minimum value at the highest daily flow ** Manual Residual were not taken



VEOLIA

InSight

Town of Ladysmith Arbutus DWTP

Monthly LRV and Turbidity Report

08/01/2024 - 09/01/2024

LRV Monthly Average

Asset	Parameter	Health	Avg	Std. Dev	Points	LL	LCL	%In	% between L and LL	% below LL	Unit
UF 1	LRV		5.0	0.0	32	4.25	4.5	100 %	0 %	0 %	#
UF 2	LRV		5.0	0.0	32	4.25	4.5	100 %	0 %	0 %	#
UF 3	LRV		5.0	0.0	32	4.25	4.5	100 %	0 %	0 %	#

LRV Daily Values

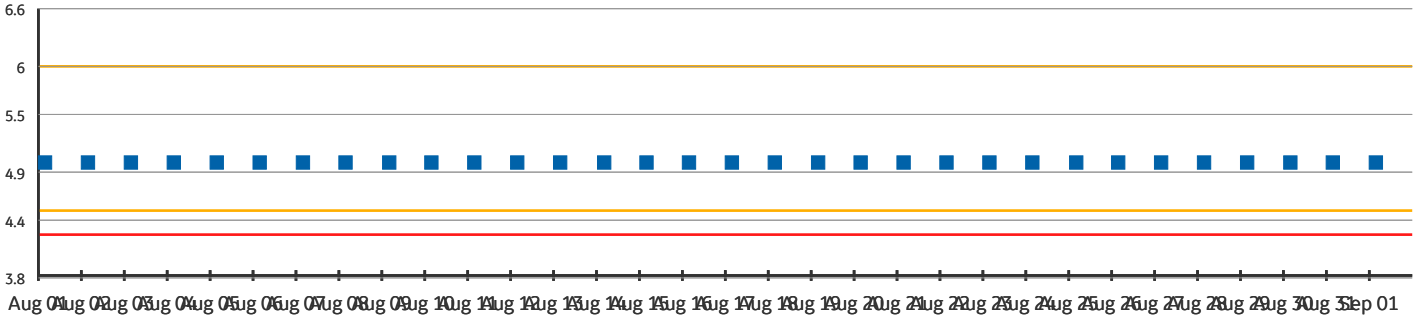
Asset	Parameter	Aug 01	Aug 02	Aug 03	Aug 04	Aug 05	Aug 06	Aug 07	Aug 08	Aug 09	Aug 10	Aug 11	Aug 12
UF 1	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 2	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 3	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0

Asset	Aug 13	Aug 14	Aug 15	Aug 16	Aug 17	Aug 18	Aug 19	Aug 20	Aug 21	Aug 22	Aug 23	Aug 24	Aug 25	Aug 26	Aug 27	Aug 28
UF 1	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 2	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 3	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0

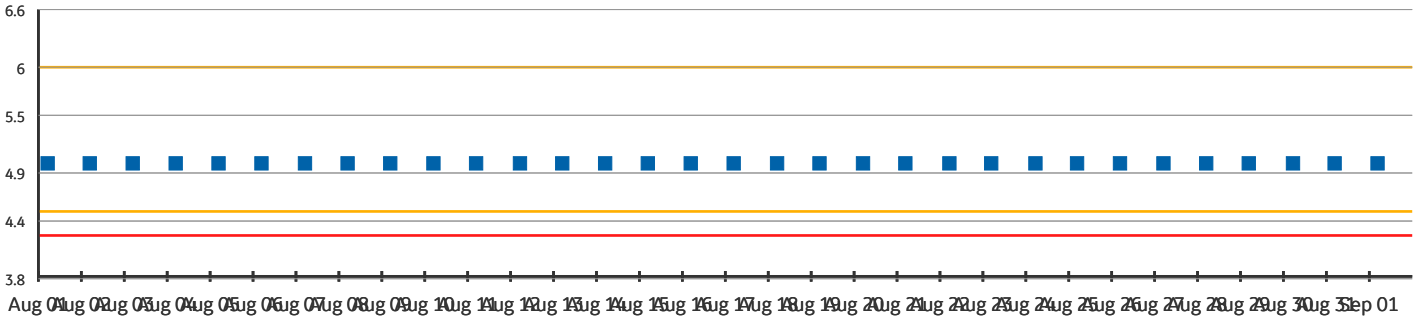
Asset	Aug 29	Aug 30	Aug 31	Sep 01
UF 1	5.0	5.0	5.0	5.0
UF 2	5.0	5.0	5.0	5.0
UF 3	5.0	5.0	5.0	5.0

LRV Raw Data

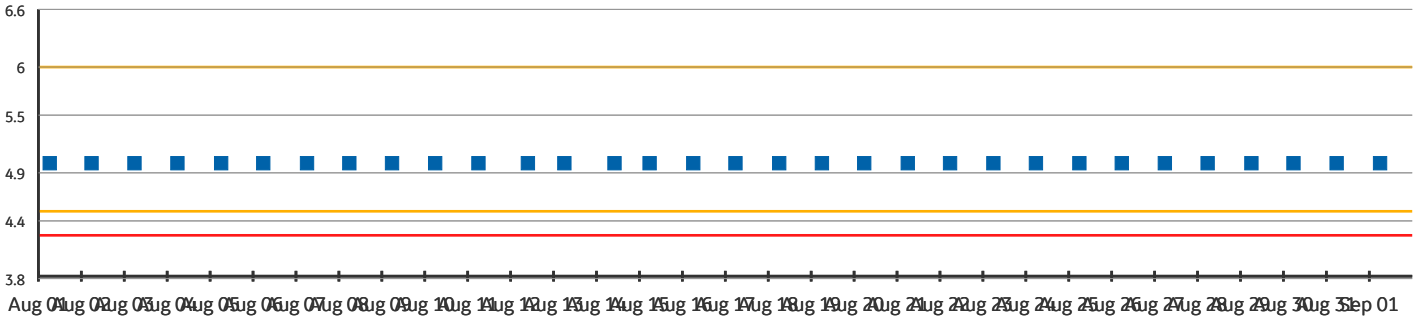
UF 1 - LRV (#)



UF 2 - LRV (#)



UF 3 - LRV (#)



Turbidity Monthly Average

Asset	Parameter	Health	Avg	Std. Dev	Points	UCL	HH	%In	% between H and HH	% above HH	Unit
UF 1	PermeateTurbidity		0.017	0.0	45205	--	--	100 %	0 %	0 %	NTU
UF 2	PermeateTurbidity		0.016	0.0	45205	--	--	100 %	0 %	0 %	NTU
UF 3	PermeateTurbidity		0.018	0.0	45205	--	--	100 %	0 %	0 %	NTU
UF 1	PermeateTurbidityAfterBP	●	0.017	0.0	530	0.1	0.3	100 %	0 %	0 %	NTU
UF 2	PermeateTurbidityAfterBP	●	0.016	0.0	529	0.1	0.3	100 %	0 %	0 %	NTU
UF 3	PermeateTurbidityAfterBP	●	0.018	0.0	533	0.1	0.3	100 %	0 %	0 %	NTU

Turbidity Daily Averages

Asset	Parameter	Aug 01	Aug 02	Aug 03	Aug 04	Aug 05	Aug 06	Aug 07	Aug 08	Aug 09	Aug 10	Aug 11
UF 1	PermeateTurbidity	0.029	0.024	0.017	0.016	0.015	0.016	0.017	0.017	0.017	0.017	0.017
UF 2	PermeateTurbidity	0.015	0.016	0.015	0.015	0.016	0.016	0.018	0.018	0.018	0.017	0.016
UF 3	PermeateTurbidity	0.019	0.019	0.019	0.019	0.019	0.02	0.021	0.021	0.021	0.02	0.02

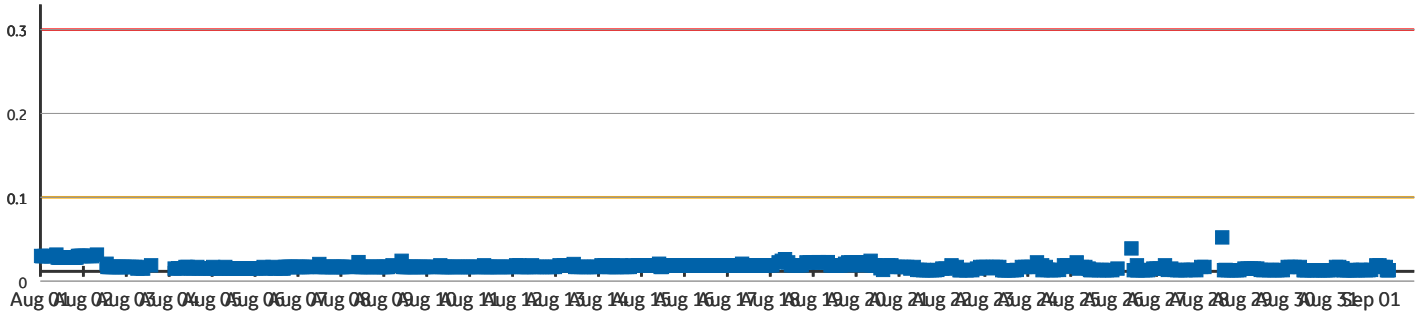
Asset	Parameter	Aug 01	Aug 02	Aug 03	Aug 04	Aug 05	Aug 06	Aug 07	Aug 08	Aug 09	Aug 10	Aug 11
UF 1	PermeateTurbidityAfterBP	0.029	0.021	0.017	0.016	0.015	0.016	0.017	0.017	0.017	0.017	0.017
UF 2	PermeateTurbidityAfterBP	0.015	0.015	0.015	0.015	0.015	0.015	0.017	0.017	0.017	0.017	0.017
UF 3	PermeateTurbidityAfterBP	0.019	0.019	0.019	0.019	0.019	0.02	0.021	0.02	0.02	0.02	0.019

Asset	Aug 12	Aug 13	Aug 14	Aug 15	Aug 16	Aug 17	Aug 18	Aug 19	Aug 20	Aug 21	Aug 22	Aug 23	Aug 24	Aug 25	Aug 26
UF 1	0.017	0.017	0.018	0.018	0.019	0.019	0.019	0.019	0.017	0.014	0.014	0.014	0.014	0.014	0.017
UF 2	0.015	0.015	0.015	0.016	0.016	0.016	0.017	0.017	0.016	0.015	0.014	0.014	0.015	0.014	0.014
UF 3	0.021	0.02	0.021	0.02	0.021	0.021	0.022	0.021	0.018	0.015	0.015	0.016	0.016	0.015	0.015
UF 1	0.017	0.018	0.018	0.019	0.019	0.019	0.021	0.021	0.019	0.014	0.015	0.015	0.016	0.015	0.016
UF 2	0.016	0.015	0.016	0.016	0.016	0.016	0.017	0.017	0.016	0.015	0.015	0.015	0.015	0.014	0.015
UF 3	0.02	0.019	0.02	0.02	0.02	0.019	0.023	0.02	0.018	0.015	0.015	0.016	0.016	0.015	0.015

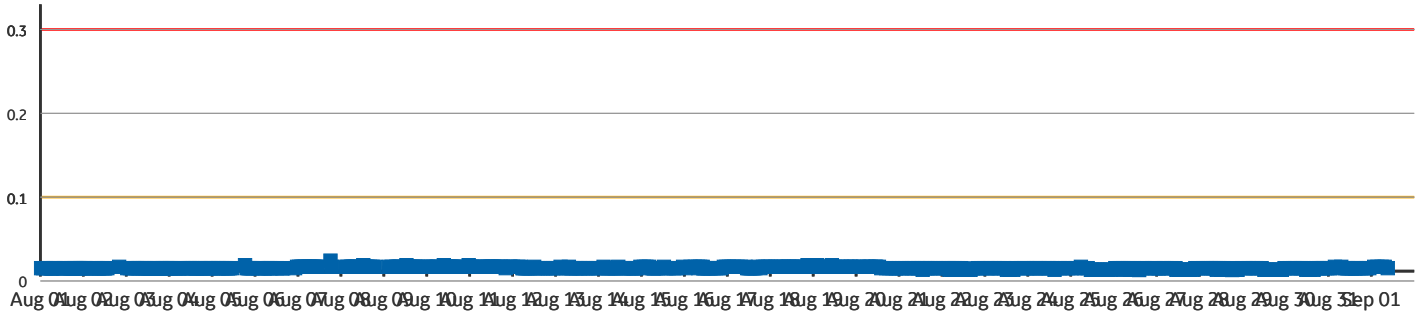
Asset	Aug 27	Aug 28	Aug 29	Aug 30	Aug 31	Sep 01
UF 1	0.013	0.019	0.013	0.014	0.014	0.014
UF 2	0.014	0.014	0.014	0.014	0.015	0.015
UF 3	0.015	0.015	0.015	0.015	0.015	0.015
UF 1	0.014	0.018	0.014	0.014	0.014	0.016
UF 2	0.014	0.015	0.014	0.015	0.015	0.017
UF 3	0.016	0.015	0.015	0.015	0.015	0.015

Turbidity Raw Data

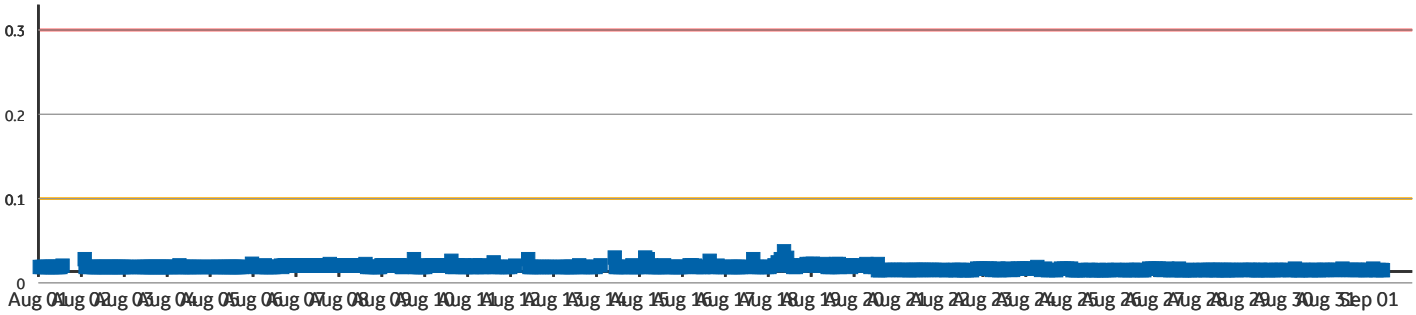
UF 1 - PermeateTurbidityAfterBP (NTU)



UF 2 - PermeateTurbidityAfterBP (NTU)



UF 3 - PermeateTurbidityAfterBP (NTU)



CERTIFICATE OF ANALYSIS

Work Order	: **VA24B9499**	Page	: 1 of 3
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 07-Aug-2024 09:45
PO	: 10880	Date Analysis Commenced	: 07-Aug-2024
C-O-C number	: ----	Issue Date	: 12-Aug-2024 10:40
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Marianne Jensen	Analyst- General	Inorganics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Inorganics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Microbiology, Burnaby, British Columbia

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances

LOR: Limit of Reporting (detection limit).

Unit	*Description*
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Analytical Results

Sub-Matrix: **Drinking Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID: Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
Client sampling date / time					06-Aug-2024 12:10	06-Aug-2024 12:20	06-Aug-2024 12:15	06-Aug-2024 12:00	----
					VA24B9499-001	VA24B9499-002	VA24B9499-003	VA24B9499-004	--------
					Result	Result	Result	Result	----
Physical Tests									
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	12.1	----
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	59.2	----
pH	----	E108/VA	0.10	pH units	----	----	----	7.40	----
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----
Organic / Inorganic Carbon									
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	1.86	0.82	0.98	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	2.01	1.00	0.76	----	----
Microbiological Tests									
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	<1	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

ALS Canada Ltd.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24B9499**	Page	: 1 of 7
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 07-Aug-2024 09:45
PO	: 10880	Issue Date	: 12-Aug-2024 10:40
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24C0379**	Page	: 1 of 4
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Weekly Sampling	Date Samples Received	: 14-Aug-2024 12:30
PO	: 10880	Date Analysis Commenced	: 14-Aug-2024
C-O-C number	: ----	Issue Date	: 20-Aug-2024 13:37
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Janice Leung	Supervisor - Organics Instrumentation	Organics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Inorganics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Microbiology, Burnaby, British Columbia
Stephanie Pinheiro	Team Leader - LCMS	LCMS, Waterloo, Ontario

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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LOR: Limit of Reporting (detection limit).

Unit	*Description*
µg/L	micrograms per litre
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID	Raw Water	UF Effluent	Treated Water (post reservoir)	Distribution system (WWTP)	----
					Client sampling date / time	13-Aug-2024 10:30	13-Aug-2024 10:30	13-Aug-2024 10:30	13-Aug-2024 10:30	----
						VA24C0379-001	VA24C0379-002	VA24C0379-003	VA24C0379-004	--------
						Result	Result	Result	Result	----
Physical Tests										
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L		----	----	8.3	----	----
Colour, true	----	E329/VA	5.0	CU		----	----	<5.0	----	----
Conductivity	----	E100/VA	2.0	µS/cm		----	----	33.2	----	----
pH	----	E108/VA	0.10	pH units		----	----	7.25	----	----
Turbidity	----	E121/VA	0.10	NTU		----	----	<0.10	----	----
Organic / Inorganic Carbon										
Carbon, dissolved inorganic [DIC]	----	E353-L/VA	0.50	mg/L		----	1.96	----	----	----
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L		2.14	2.04	----	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L		2.41	2.16	----	----	----
Microbiological Tests										
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL		----	----	<1	----	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL		----	----	<1	----	----
Coliforms, total	----	E010/VA	1	MPN/100mL		----	----	<1	----	----
Volatile Organic Compounds [THMs]										
Bromodichloromethane	75-27-4	E611B/VA	1.0	µg/L		----	----	2.7	3.2	----
Bromoform	75-25-2	E611B/VA	1.0	µg/L		----	----	<1.0	<1.0	----
Chloroform	67-66-3	E611B/VA	1.0	µg/L		----	----	70.6	83.9	----
Dibromochloromethane	124-48-1	E611B/VA	1.0	µg/L		----	----	<1.0	<1.0	----
Trihalomethanes [THMs], total	----	E611B/VA	2.0	µg/L		----	----	73.3	87.1	----
Volatile Organic Compounds [THMs] Surrogates										
Bromofluorobenzene, 4-	460-00-4	E611B/VA	1.0	%		----	----	94.2	98.4	----
Difluorobenzene, 1,4-	540-36-3	E611B/VA	1.0	%		----	----	101	100.0	----
Haloacetic Acids										
Bromochloroacetic acid	5589-96-8	E750/WT	1.00	µg/L		----	----	<1.00	<1.00	----
Dibromoacetic acid	631-64-1	E750/WT	1.00	µg/L		----	----	<1.00	<1.00	----
Dichloroacetic acid	79-43-6	E750/WT	1.00	µg/L		----	----	26.3	30.3	----
Monobromoacetic acid	79-08-3	E750/WT	1.00	µg/L		----	----	<1.00	<1.00	----
Monochloroacetic acid	79-11-8	E750/WT	1.00	µg/L		----	----	1.41	1.67	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Raw Water	UF Effluent	Treated Water (post reservoir)	Distribution system (WWTP)	----
					Client sampling date / time	13-Aug-2024 10:30	13-Aug-2024 10:30	13-Aug-2024 10:30	13-Aug-2024 10:30	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24C0379-001	VA24C0379-002	VA24C0379-003	VA24C0379-004	--------
						Result	Result	Result	Result	----
Haloacetic Acids										
Trichloroacetic acid	76-03-9	E750/WT	1.00	µg/L		----	----	38.8	45.9	----
Haloacetic acids, total [HAA5]	n/a	E750/WT	5.00	µg/L		----	----	66.5	77.9	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24C0379**	Page	: 1 of 9
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Weekly Sampling	Date Samples Received	: 14-Aug-2024 12:30
PO	: 10880	Issue Date	: 20-Aug-2024 13:36
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Matrix Spike Duplicate (MSD) outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- Quality Control Sample Frequency Outliers occur - please see following pages for full details.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24C1216**	Page	: 1 of 4
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 21-Aug-2024 12:05
PO	: PO #10916	Date Analysis Commenced	: 21-Aug-2024
C-O-C number	: 17-Week 2	Issue Date	: 03-Sep-2024 09:38
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Janice Leung	Supervisor - Organics Instrumentation	Organics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Inorganics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Microbiology, Burnaby, British Columbia
Stephanie Pinheiro	Team Leader - LCMS	LCMS, Waterloo, Ontario
Tracy Harley	Supervisor - Water Quality Instrumentation	Inorganics, Burnaby, British Columbia

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances

LOR: Limit of Reporting (detection limit).

Unit	*Description*
µg/L	micrograms per litre
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Workorder Comments

Treated water sample :Exceeded Recommended Holding Time prior to receipt at the lab for HPC analysis.Requested HPC analysis will be carried out from the expired sample.

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Raw Water	UF Effluent	Treated Water (post reservoir)	Distribution System (WWTP)	----
Client sample ID					VA24C1216-001	VA24C1216-002	VA24C1216-003	VA24C1216-004	--------
Client sampling date / time					20-Aug-2024 10:30	20-Aug-2024 10:30	20-Aug-2024 10:30	20-Aug-2024 10:30	----
					Result	Result	Result	Result	----
Physical Tests									
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	10.2	----	----
Colour, true	----	E329/VA	5.0	CU	----	----	<5.0	----	----
Conductivity	----	E100/VA	2.0	µS/cm	----	----	38.6	----	----
pH	----	E108/VA	0.10	pH units	----	----	7.32	----	----
Turbidity	----	E121/VA	0.10	NTU	----	----	<0.10	----	----
Organic / Inorganic Carbon									
Carbon, dissolved inorganic [DIC]	----	E353-L/VA	0.50	mg/L	----	2.25	----	----	----
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	2.05	2.11	----	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	2.20	2.26	----	----	----
Microbiological Tests									
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	<1	----	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	<1	----	----
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	<1	----	----
Volatile Organic Compounds [THMs]									
Bromodichloromethane	75-27-4	E611B/VA	1.0	µg/L	----	----	2.9	3.4	----
Bromoform	75-25-2	E611B/VA	1.0	µg/L	----	----	<1.0	<1.0	----
Chloroform	67-66-3	E611B/VA	1.0	µg/L	----	----	84.7	107	----
Dibromochloromethane	124-48-1	E611B/VA	1.0	µg/L	----	----	<1.0	<1.0	----
Trihalomethanes [THMs], total	----	E611B/VA	2.0	µg/L	----	----	87.6	110	----
Volatile Organic Compounds [THMs] Surrogates									
Bromofluorobenzene, 4-	460-00-4	E611B/VA	1.0	%	----	----	94.5	91.6	----
Difluorobenzene, 1,4-	540-36-3	E611B/VA	1.0	%	----	----	101	99.3	----
Haloacetic Acids									
Bromochloroacetic acid	5589-96-8	E750/WT	1.00	µg/L	----	----	<1.00	1.04	----
Dibromoacetic acid	631-64-1	E750/WT	1.00	µg/L	----	----	<1.00	<1.00	----
Dichloroacetic acid	79-43-6	E750/WT	1.00	µg/L	----	----	34.1	37.6	----
Monobromoacetic acid	79-08-3	E750/WT	1.00	µg/L	----	----	<1.00	<1.00	----
Monochloroacetic acid	79-11-8	E750/WT	1.00	µg/L	----	----	1.55	2.07	----

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID	Raw Water	UF Effluent	Treated Water (post reservoir)	Distribution System (WWTP)	----
					Client sampling date / time	20-Aug-2024 10:30	20-Aug-2024 10:30	20-Aug-2024 10:30	20-Aug-2024 10:30	----
						VA24C1216-001	VA24C1216-002	VA24C1216-003	VA24C1216-004	--------
						Result	Result	Result	Result	----
Haloacetic Acids										
Trichloroacetic acid	76-03-9	E750/WT	1.00	µg/L		----	----	36.7	43.1	----
Haloacetic acids, total [HAA5]	n/a	E750/WT	5.00	µg/L		----	----	72.4	82.8	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24C1216**	Page	: 1 of 9
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 21-Aug-2024 12:05
PO	: PO #10916	Issue Date	: 03-Sep-2024 09:34
C-O-C number	: 17-Week 2		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

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CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

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Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Matrix Spike Duplicate (MSD) outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24C1486**	Page	: 1 of 6
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Quarterly Lake Sampling	Date Samples Received	: 22-Aug-2024 12:20
PO	: 10880	Date Analysis Commenced	: 23-Aug-2024
C-O-C number	: ----	Issue Date	: 03-Sep-2024 16:00
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 2		
No. of samples analysed	: 2		

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- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Janice Leung	Supervisor - Organics Instrumentation	Organics, Burnaby, British Columbia
Kevin Duarte	Supervisor - Metals ICP Instrumentation	Metals, Burnaby, British Columbia
Lindsay Gung	Supervisor - Water Chemistry	Inorganics, Burnaby, British Columbia
Maya Urquhart	Lab Analyst	Metals, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Microbiology, Burnaby, British Columbia

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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LOR: Limit of Reporting (detection limit).

Unit	*Description*
% T/cm	% transmittance per centimetre
µg/L	micrograms per litre
AU/cm	absorbance units per centimetre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Analytical Results

Sub-Matrix: Water
(Matrix: Water)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID: Stocking Lake	Holland Lake	----	----	----
Client sampling date / time					21-Aug-2024 11:00	21-Aug-2024 09:30	----	----	----
					VA24C1486-001	VA24C1486-002	--------	--------	--------
					Result	Result	----	----	----
Physical Tests									
Absorbance, UV (@ 254nm)	----	E404/VA	0.0050	AU/cm	0.0580	0.0740	----	----	----
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	9.8	4.3	----	----	----
Colour, true	----	E329/VA	5.0	CU	<5.0	<5.0	----	----	----
Hardness (as CaCO3), from total Ca/Mg	----	EC100A/VA	0.60	mg/L	10.6	4.75	----	----	----
pH	----	E108/VA	0.10	pH units	7.13	6.73	----	----	----
Solids, total dissolved [TDS]	----	E162/VA	10	mg/L	28	23	----	----	----
Turbidity	----	E121/VA	0.10	NTU	0.47	0.38	----	----	----
Transmittance, UV (@ 254nm)	----	E404/VA	1.0	% T/cm	87.5	84.3	----	----	----
Anions and Nutrients									
Bromide	24959-67-9	E235.Br-L/VA	0.050	mg/L	<0.050	<0.050	----	----	----
Organic / Inorganic Carbon									
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	2.43	3.18	----	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	2.78	3.37	----	----	----
Microbiological Tests									
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	<1	<1	----	----	----
Coliforms, total	----	E010/VA	1	MPN/100mL	1200	1410	----	----	----
Total Metals									
Aluminum, total	7429-90-5	E420/VA	0.0030	mg/L	0.0300	0.0320	----	----	----
Antimony, total	7440-36-0	E420/VA	0.00010	mg/L	<0.00010	<0.00010	----	----	----
Arsenic, total	7440-38-2	E420/VA	0.00010	mg/L	0.00013	<0.00010	----	----	----
Barium, total	7440-39-3	E420/VA	0.00010	mg/L	0.00307	0.00299	----	----	----
Beryllium, total	7440-41-7	E420/VA	0.000100	mg/L	<0.000100	<0.000100	----	----	----
Bismuth, total	7440-69-9	E420/VA	0.000050	mg/L	<0.000050	<0.000050	----	----	----
Boron, total	7440-42-8	E420/VA	0.010	mg/L	0.022	<0.010	----	----	----
Cadmium, total	7440-43-9	E420/VA	0.0000050	mg/L	<0.0000050	<0.0000050	----	----	----
Calcium, total	7440-70-2	E420/VA	0.050	mg/L	3.34	1.46	----	----	----
Cesium, total	7440-46-2	E420/VA	0.000010	mg/L	0.000011	<0.000010	----	----	----
Chromium, total	7440-47-3	E420/VA	0.00050	mg/L	<0.00050	<0.00050	----	----	----
Cobalt, total	7440-48-4	E420/VA	0.00010	mg/L	<0.00010	<0.00010	----	----	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Stocking Lake	Holland Lake	----	----	----
					Client sampling date / time	21-Aug-2024 11:00	21-Aug-2024 09:30	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24C1486-001	VA24C1486-002	--------	--------	--------
						Result	Result	----	----	----
Total Metals										
Copper, total	7440-50-8	E420/VA	0.00050	mg/L		0.00054	0.00085	----	----	----
Iron, total	7439-89-6	E420/VA	0.010	mg/L		0.050	0.115	----	----	----
Lead, total	7439-92-1	E420/VA	0.000050	mg/L		<0.000050	<0.000050	----	----	----
Lithium, total	7439-93-2	E420/VA	0.0010	mg/L		<0.0010	<0.0010	----	----	----
Magnesium, total	7439-95-4	E420/VA	0.0050	mg/L		0.559	0.269	----	----	----
Manganese, total	7439-96-5	E420/VA	0.00010	mg/L		0.00426	0.0131	----	----	----
Mercury, total	7439-97-6	E508/VA	0.0000050	mg/L		<0.0000050	<0.0000050	----	----	----
Molybdenum, total	7439-98-7	E420/VA	0.000050	mg/L		0.000236	<0.000050	----	----	----
Nickel, total	7440-02-0	E420/VA	0.00050	mg/L		<0.00050	<0.00050	----	----	----
Phosphorus, total	7723-14-0	E420/VA	0.050	mg/L		<0.050	<0.050	----	----	----
Potassium, total	7440-09-7	E420/VA	0.050	mg/L		0.279	0.169	----	----	----
Rubidium, total	7440-17-7	E420/VA	0.00020	mg/L		0.00067	0.00053	----	----	----
Selenium, total	7782-49-2	E420/VA	0.000050	mg/L		<0.000050	0.000054	----	----	----
Silicon, total	7440-21-3	E420/VA	0.10	mg/L		1.83	1.00	----	----	----
Silver, total	7440-22-4	E420/VA	0.000010	mg/L		<0.000010	<0.000010	----	----	----
Sodium, total	7440-23-5	E420/VA	0.050	mg/L		1.20	0.761	----	----	----
Strontium, total	7440-24-6	E420/VA	0.00020	mg/L		0.0113	0.00710	----	----	----
Sulfur, total	7704-34-9	E420/VA	0.50	mg/L		0.54	<0.50	----	----	----
Tellurium, total	13494-80-9	E420/VA	0.00020	mg/L		<0.00020	<0.00020	----	----	----
Thallium, total	7440-28-0	E420/VA	0.000010	mg/L		<0.000010	<0.000010	----	----	----
Thorium, total	7440-29-1	E420/VA	0.00010	mg/L		<0.00010	<0.00010	----	----	----
Tin, total	7440-31-5	E420/VA	0.00010	mg/L		<0.00010	<0.00010	----	----	----
Titanium, total	7440-32-6	E420/VA	0.00030	mg/L		<0.00030	<0.00030	----	----	----
Tungsten, total	7440-33-7	E420/VA	0.00010	mg/L		<0.00010	<0.00010	----	----	----
Uranium, total	7440-61-1	E420/VA	0.000010	mg/L		0.000019	<0.000010	----	----	----
Vanadium, total	7440-62-2	E420/VA	0.00050	mg/L		<0.00050	<0.00050	----	----	----
Zinc, total	7440-66-6	E420/VA	0.0030	mg/L		<0.0030	<0.0030	----	----	----
Zirconium, total	7440-67-7	E420/VA	0.00020	mg/L		<0.00020	<0.00020	----	----	----
Hydrocarbons										
EPH (C10-C19)	----	E601A/VA	250	µg/L		<250	<250	----	----	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Stocking Lake	Holland Lake	----	----	----
					Client sampling date / time	21-Aug-2024 11:00	21-Aug-2024 09:30	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24C1486-001	VA24C1486-002	--------	--------	--------
						Result	Result	----	----	----
Hydrocarbons										
EPH (C10-C32)	----	E601A/VA	400	µg/L		<400	<400	----	----	----
EPH (C19-C32)	----	E601A/VA	250	µg/L		<250	<250	----	----	----
TEH (C10-C30), BC	----	E601A/VA	250	µg/L		<250	<250	----	----	----
Hydrocarbons Surrogates										
Bromobenzotrifluoride, 2- (EPH surrogate)	392-83-6	E601A/VA	1.0	%		91.4	91.9	----	----	----
Polycyclic Aromatic Hydrocarbons										
Acenaphthene	83-32-9	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Acenaphthylene	208-96-8	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Acridine	260-94-6	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Anthracene	120-12-7	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Benz(a)anthracene	56-55-3	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Benzo(a)pyrene	50-32-8	E641A/VA	0.0050	µg/L		<0.0050	<0.0050	----	----	----
Benzo(b+j)fluoranthene	n/a	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Benzo(b+j+k)fluoranthene	n/a	E641A/VA	0.015	µg/L		<0.015	<0.015	----	----	----
Benzo(g,h,i)perylene	191-24-2	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Benzo(k)fluoranthene	207-08-9	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Chrysene	218-01-9	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Dibenz(a,h)anthracene	53-70-3	E641A/VA	0.0050	µg/L		<0.0050	<0.0050	----	----	----
Fluoranthene	206-44-0	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Fluorene	86-73-7	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Indeno(1,2,3-c,d)pyrene	193-39-5	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Methylnaphthalene, 1-	90-12-0	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Methylnaphthalene, 2-	91-57-6	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Naphthalene	91-20-3	E641A/VA	0.050	µg/L		<0.050	<0.050	----	----	----
Phenanthrene	85-01-8	E641A/VA	0.020	µg/L		<0.020	<0.020	----	----	----
Pyrene	129-00-0	E641A/VA	0.010	µg/L		<0.010	<0.010	----	----	----
Quinoline	91-22-5	E641A/VA	0.050	µg/L		<0.050	<0.050	----	----	----
Polycyclic Aromatic Hydrocarbons Surrogates										
Chrysene-d12	1719-03-5	E641A/VA	0.1	%		102	100	----	----	----
Naphthalene-d8	1146-65-2	E641A/VA	0.1	%		95.4	96.3	----	----	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Stocking Lake	Holland Lake	----	----	----
					Client sampling date / time	21-Aug-2024 11:00	21-Aug-2024 09:30	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24C1486-001	VA24C1486-002	--------	--------	--------
						Result	Result	----	----	----
Polycyclic Aromatic Hydrocarbons Surrogates										
Phenanthrene-d10	1517-22-2	E641A/VA	0.1	%		106	107	----	----	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24C1486**	Page	: 1 of 11
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Quarterly Lake Sampling	Date Samples Received	: 22-Aug-2024 12:20
PO	: 10880	Issue Date	: 03-Sep-2024 16:27
C-O-C number	: ----		
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 2		
No. of samples analysed	: 2		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24C2138**	Page	: 1 of 4
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Weekly Sampling	Date Samples Received	: 28-Aug-2024 11:00
PO	: 10880	Date Analysis Commenced	: 28-Aug-2024
C-O-C number	: 17-Week 3	Issue Date	: 09-Sep-2024 10:56
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Lindsay Gung	Supervisor - Water Chemistry	Microbiology, Burnaby, British Columbia
Lindsay Gung	Supervisor - Water Chemistry	Organics, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Inorganics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Inorganics, Burnaby, British Columbia
Stephanie Pinheiro	Team Leader - LCMS	LCMS, Waterloo, Ontario

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances

LOR: Limit of Reporting (detection limit).

Unit	*Description*
µg/L	micrograms per litre
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID	Raw Water	UF Effluent	Treated Water (post reservoir)	Distribution System (WWTP)	----
					Client sampling date / time	27-Aug-2024 10:30	27-Aug-2024 10:30	27-Aug-2024 10:30	27-Aug-2024 10:30	----
						VA24C2138-001	VA24C2138-002	VA24C2138-003	VA24C2138-004	--------
						Result	Result	Result	Result	----
Physical Tests										
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L		----	----	10.0	----	----
Colour, true	----	E329/VA	5.0	CU		----	----	<5.0	----	----
Conductivity	----	E100/VA	2.0	µS/cm		----	----	36.0	----	----
pH	----	E108/VA	0.10	pH units		----	----	7.28	----	----
Turbidity	----	E121/VA	0.10	NTU		----	----	<0.10	----	----
Organic / Inorganic Carbon										
Carbon, dissolved inorganic [DIC]	----	E353-L/VA	0.50	mg/L		----	2.20	----	----	----
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L		1.71	1.79	----	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L		2.79	2.11	----	----	----
Microbiological Tests										
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL		----	----	<1	----	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL		----	----	<1	----	----
Coliforms, total	----	E010/VA	1	MPN/100mL		----	----	<1	----	----
Volatile Organic Compounds [THMs]										
Bromodichloromethane	75-27-4	E611B/VA	1.0	µg/L		----	----	3.2	4.3	----
Bromoform	75-25-2	E611B/VA	1.0	µg/L		----	----	<1.0	<1.0	----
Chloroform	67-66-3	E611B/VA	1.0	µg/L		----	----	78.7	120	----
Dibromochloromethane	124-48-1	E611B/VA	1.0	µg/L		----	----	<1.0	<1.0	----
Trihalomethanes [THMs], total	----	E611B/VA	2.0	µg/L		----	----	81.9	124	----
Volatile Organic Compounds [THMs] Surrogates										
Bromofluorobenzene, 4-	460-00-4	E611B/VA	1.0	%		----	----	93.8	94.7	----
Difluorobenzene, 1,4-	540-36-3	E611B/VA	1.0	%		----	----	98.7	98.3	----
Haloacetic Acids										
Bromochloroacetic acid	5589-96-8	E750/WT	1.00	µg/L		----	----	<1.00	1.01	----
Dibromoacetic acid	631-64-1	E750/WT	1.00	µg/L		----	----	<1.00	<1.00	----
Dichloroacetic acid	79-43-6	E750/WT	1.00	µg/L		----	----	26.5	36.3	----
Monobromoacetic acid	79-08-3	E750/WT	1.00	µg/L		----	----	<1.00	<1.00	----
Monochloroacetic acid	79-11-8	E750/WT	1.00	µg/L		----	----	1.46	2.27	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Raw Water	UF Effluent	Treated Water (post reservoir)	Distribution System (WWTP)	----
					Client sampling date / time	27-Aug-2024 10:30	27-Aug-2024 10:30	27-Aug-2024 10:30	27-Aug-2024 10:30	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24C2138-001	VA24C2138-002	VA24C2138-003	VA24C2138-004	--------
						Result	Result	Result	Result	----
Haloacetic Acids										
Trichloroacetic acid	76-03-9	E750/WT	1.00	µg/L		----	----	30.5	37.2	----
Haloacetic acids, total [HAA5]	n/a	E750/WT	5.00	µg/L		----	----	58.5	75.8	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24C2138**	Page	: 1 of 9
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Weekly Sampling	Date Samples Received	: 28-Aug-2024 11:00
PO	: 10880	Issue Date	: 09-Sep-2024 10:56
C-O-C number	: 17-Week 3		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

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Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Matrix Spike Duplicate (MSD) outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24C2590**	Page	: 1 of 4
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Quarterly DT Sampling	Date Samples Received	: 30-Aug-2024 10:50
PO	: 10880	Date Analysis Commenced	: 31-Aug-2024
C-O-C number	: ----	Issue Date	: 11-Sep-2024 16:39
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Ghazaleh Khanmirzaei	Analyst	Metals, Burnaby, British Columbia
Kate Dimitrova	Supervisor - Inorganic	Inorganics, Burnaby, British Columbia
Kim Jensen	Department Manager - Metals	Metals, Burnaby, British Columbia

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances

LOR: Limit of Reporting (detection limit).

Unit	*Description*
mg/L	milligrams per litre
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID	FJCC	Town Hall	Fire Department	RCMP	----
					Client sampling date / time	29-Aug-2024 10:30	29-Aug-2024 10:30	29-Aug-2024 10:30	29-Aug-2024 10:30	----
						VA24C2590-001	VA24C2590-002	VA24C2590-003	VA24C2590-004	--------
						Result	Result	Result	Result	----
Physical Tests										
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L		10.0	10.1	10.2	11.5	----
Hardness (as CaCO3), from total Ca/Mg	----	EC100A/VA	0.60	mg/L		6.86	7.26	6.92	7.69	----
pH	----	E108/VA	0.10	pH units		7.33	7.35	7.36	7.44	----
Solids, total dissolved [TDS]	----	E162/VA	10	mg/L		33	31	29	28	----
Total Metals										
Aluminum, total	7429-90-5	E420/VA	0.0030	mg/L		0.0293	0.0231	0.0162	0.0438	----
Antimony, total	7440-36-0	E420/VA	0.00010	mg/L		<0.00010	<0.00010	<0.00010	<0.00010	----
Arsenic, total	7440-38-2	E420/VA	0.00010	mg/L		0.00014	0.00010	<0.00010	<0.00010	----
Barium, total	7440-39-3	E420/VA	0.00010	mg/L		0.00330	0.00323	0.00308	0.00265	----
Beryllium, total	7440-41-7	E420/VA	0.000100	mg/L		<0.000100	<0.000100	<0.000100	<0.000100	----
Bismuth, total	7440-69-9	E420/VA	0.000050	mg/L		<0.000050	<0.000050	<0.000050	<0.000050	----
Boron, total	7440-42-8	E420/VA	0.010	mg/L		0.022	0.022	0.022	0.024	----
Cadmium, total	7440-43-9	E420/VA	0.0000050	mg/L		<0.0000050	<0.0000050	<0.0000050	<0.0000050	----
Calcium, total	7440-70-2	E420/VA	0.050	mg/L		2.18	2.26	2.18	2.24	----
Cesium, total	7440-46-2	E420/VA	0.000010	mg/L		<0.000010	<0.000010	<0.000010	<0.000010	----
Chromium, total	7440-47-3	E420/VA	0.00050	mg/L		<0.00050	<0.00050	<0.00050	<0.00050	----
Cobalt, total	7440-48-4	E420/VA	0.00010	mg/L		<0.00010	<0.00010	<0.00010	<0.00010	----
Copper, total	7440-50-8	E420/VA	0.00050	mg/L		0.0192	0.0404	0.229	0.241	----
Iron, total	7439-89-6	E420/VA	0.010	mg/L		0.021	<0.010	<0.010	0.083	----
Lead, total	7439-92-1	E420/VA	0.000050	mg/L		0.000736	0.000186	0.00182	0.00105	----
Lithium, total	7439-93-2	E420/VA	0.0010	mg/L		<0.0010	<0.0010	<0.0010	<0.0010	----
Magnesium, total	7439-95-4	E420/VA	0.0050	mg/L		0.344	0.392	0.359	0.510	----
Manganese, total	7439-96-5	E420/VA	0.00010	mg/L		0.00037	0.00016	0.00038	0.00125	----
Mercury, total	7439-97-6	E508/VA	0.0000050	mg/L		<0.0000050	<0.0000050	<0.0000050	<0.0000050	----
Molybdenum, total	7439-98-7	E420/VA	0.000050	mg/L		0.000102	0.000117	0.000084	0.000118	----
Nickel, total	7440-02-0	E420/VA	0.00050	mg/L		<0.00050	<0.00050	<0.00050	<0.00050	----
Phosphorus, total	7723-14-0	E420/VA	0.050	mg/L		<0.050	<0.050	<0.050	<0.050	----
Potassium, total	7440-09-7	E420/VA	0.050	mg/L		0.161	0.159	0.164	0.168	----
Rubidium, total	7440-17-7	E420/VA	0.00020	mg/L		0.00047	0.00046	0.00049	0.00054	----

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID	FJCC	Town Hall	Fire Department	RCMP	----
					Client sampling date / time	29-Aug-2024 10:30	29-Aug-2024 10:30	29-Aug-2024 10:30	29-Aug-2024 10:30	----
						VA24C2590-001	VA24C2590-002	VA24C2590-003	VA24C2590-004	--------
						Result	Result	Result	Result	----
Total Metals										
Selenium, total	7782-49-2	E420/VA	0.000050	mg/L		<0.000050	<0.000050	<0.000050	<0.000050	----
Silicon, total	7440-21-3	E420/VA	0.10	mg/L		1.83	1.84	1.68	1.76	----
Silver, total	7440-22-4	E420/VA	0.000010	mg/L		<0.000010	<0.000010	0.000016	<0.000010	----
Sodium, total	7440-23-5	E420/VA	0.050	mg/L		4.37	4.49	4.32	4.56	----
Strontium, total	7440-24-6	E420/VA	0.00020	mg/L		0.0102	0.0106	0.00996	0.0102	----
Sulfur, total	7704-34-9	E420/VA	0.50	mg/L		<0.50	<0.50	<0.50	<0.50	----
Tellurium, total	13494-80-9	E420/VA	0.00020	mg/L		<0.00020	<0.00020	<0.00020	<0.00020	----
Thallium, total	7440-28-0	E420/VA	0.000010	mg/L		<0.000010	<0.000010	<0.000010	<0.000010	----
Thorium, total	7440-29-1	E420/VA	0.00010	mg/L		<0.00010	<0.00010	<0.00010	<0.00010	----
Tin, total	7440-31-5	E420/VA	0.00010	mg/L		<0.00010	<0.00010	<0.00010	<0.00010	----
Titanium, total	7440-32-6	E420/VA	0.00030	mg/L		<0.00030	<0.00030	<0.00030	<0.00030	----
Tungsten, total	7440-33-7	E420/VA	0.00010	mg/L		<0.00010	<0.00010	<0.00010	<0.00010	----
Uranium, total	7440-61-1	E420/VA	0.000010	mg/L		0.000011	0.000011	<0.000010	<0.000010	----
Vanadium, total	7440-62-2	E420/VA	0.00050	mg/L		<0.00050	<0.00050	<0.00050	<0.00050	----
Zinc, total	7440-66-6	E420/VA	0.0030	mg/L		<0.0030	<0.0030	0.0057	0.0074	----
Zirconium, total	7440-67-7	E420/VA	0.00020	mg/L		<0.00020	<0.00020	<0.00020	<0.00020	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24C2590**	Page	: 1 of 8
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Quarterly DT Sampling	Date Samples Received	: 30-Aug-2024 10:50
PO	: 10880	Issue Date	: 11-Sep-2024 16:38
C-O-C number	: ----		
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- Matrix Spike outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.